

```
Processando config 13, seed 4, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin
```

```
smultin_MultinomialBitsOver test:
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```
-----
N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE
```

```
Number of bits = n = 8388608
Number of cells =  $2^L$  = 274877906944
Expected number per cell =  $1 / 32768$ 
EColl =  $n^2 / (2k)$  = 128
Hashing = TRUE
```

```
Collision test
CollisionOver: density =  $n / k = 1 / 32768.00$ 
Expected number of collisions =  $\mu = 128.00$ 
```

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Results of CollisionOver test:
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POISSON approximation :
Expected number of collisions =  $N \cdot \mu$  : 256.00
Observed number of collisions : 326
p-value of test : 1.5e-5 *****
```

```
-----
Total number of cells containing j balls
```

```
j = 0 : 549739036998
j = 1 : 16776564
j = 2 : 326
j = 3 : 0
j = 4 : 0
j = 5 : 0
```

```
-----
CPU time used : 00:00:04.51
```

```
Generator state:
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin
```

snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.20
p-value of test : 0.42

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 3.10
p-value of test : 0.88

Normal statistic for number of jumps : 0.077
p-value of test : 0.47

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.24
p-value of test : 0.41

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.46
p-value of test : 0.68

CPU time used : 00:00:00.19

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.023
p-value of test : 0.57

Kolmogorov-Smirnov- statistic = D- : 0.027
p-value of test : 0.47

Anderson-Darling statistic = A2 : 0.86
p-value of test : 0.44

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 6.67
p-value of test : 0.08

Global longest run of 1 : 25.00
p-value of test : 0.22

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 109.44
p-value of test : 0.13

CPU time used : 00:00:00.01

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 27.23
p-value of test : 0.20

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.69
p-value of test : 0.95

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.14
p-value of test : 0.44

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : -1.71
p-value of test : 0.96

CPU time used : 00:00:00.01

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 167.39
p-value of test : 0.69

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 310.17
p-value of test : 0.41

CPU time used : 00:00:00.01

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 510.33
p-value of test : 0.24

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

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Normal statistic          :   -0.29
p-value of test          :    0.61
-----
```

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CPU time used            :   00:00:00.00
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Generator state:
 269499904 bits have been read.
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :   -0.042
p-value of test          :    0.52
-----
```

```
CPU time used            :   00:00:00.01
```

```
Generator state:
 286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin
```

```
sstring_Run test:
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```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355444
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :   46.05
p-value of test                   :    0.12
-----
```

```
-----
Total number of bits:  13420608
```

```
Normal statistic for number of bits :   -0.22
p-value of test                   :    0.59
```

CPU time used : 00:00:00.05

Generator state:
299697696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 1.25
p-value of test : 0.74

CPU time used : 00:00:00.06

Generator state:
316474912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.08

Generator state:
333166112 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	48.51
p-value of test	:	0.23

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	62.45
p-value of test	:	0.02

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	67.21
p-value of test	:	0.37

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	33.59
p-value of test	:	0.58

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	27.08
p-value of test	:	0.21

CPU time used : 00:00:00.05

Generator state:

349943328 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	104.28
p-value of test	:	0.22

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	89.50
p-value of test	:	0.61

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	531.27
p-value of test	:	0.27

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	104.66
p-value of test	:	0.14

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	50.32
p-value of test	:	0.46

CPU time used : 00:00:00.05

Generator state:

366720544 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config13_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	131.51
p-value of test	:	0.45

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	136.03
p-value of test	:	0.36

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	131.96
p-value of test	:	0.95

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	129.18
p-value of test	:	0.50

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	82.75
p-value of test	:	0.67

CPU time used : 00:00:00.04

Generator state:

383497344 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.17
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
1 MultinomialBitsOver	1.5e-5
22 MatrixRank, 320 x 320	eps
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All other tests were passed